Model

Item	Value
Architecture	two-head Conformer-S (intent + shared slot head)
Parameters / weights	**246,055** params · **0.99 MB** ckpt
Deployed artifact	ONNX dual-output: `intent_logits [B,20]` + `slot_logits [B,19]`
Float / INT8 size	1.03 MB / **0.32 MB** (int8, 98.4% argmax parity)
Test accuracy	**85.2%** [84.2, 86.2] · balanced acc 83.5%
Slot exact match	**69.3%** end-to-end · 77.3% given correct intent
Wake word	openWakeWord *"hey mycroft"* (peak 0.9996, 0/250 FA @0.75)

Validation on the Raspberry Pi 4/5

Item	Value
Keyword / intent acc	85.2% / 83.5% (held-out, unseen speakers)
False-accept rate	0.8% @0.5 → **0% @0.75** (250 negatives)
Latency p95 / RTF	0.88 ms / 0.0008 (Apple-Silicon CPU) · **TBD-on-Pi** on Pi
Runtime	onnxruntime, 1 thread, INT8, fully offline

Dataset & Training

Dataset

Item	Value
Source	HF `airimonda/ai231-me2-voice-commands` · DOI 10.57967/hf/10723
Hours / utts	9.6 h / 16,628 clips (10,733 train · 4,443 test · 1,250 neg · 202 holdout)
Speakers	864 (train 307 / test 115 / holdout 5 — *fully disjoint*)
Labels	20 intents · 18 slot values (6 slotted intents)

Training (Apple-Silicon MPS — no A100 cluster available)

Item	Value
Cluster	local M-series (MPS), 1 GPU-equiv · seed 0
Objective	joint CE: intent (20-way) + slot (19-way)
Optimiser	AdamW, lr 3e-4, wd 1e-4, label smoothing 0.1
Steps / loss	84 epochs (early-stop) · train loss 2.15 · 39.3 min
Mic · 16 kHz → log-mel	40 × 100 (80 ms hop, 1 s window)
Encoder	1 Conformer layer · 64-dim · 4 heads (non-causal, offline clip)
Heads	intent head (20) · shared slot head (19)

Reviewer Checklist

#	Item	Status
1	Repo public, one-command reproduction	**partial** — `setup_pi.sh` + `-demo` one-command; repo to be pushed public
2	Dataset licensed and citable (DOI)	**done** — HF DOI 10.57967/hf/10723
3	Training logs + final checkpoint committed	**done** — `train_log.json`, `history.csv`, `best.pt` for all 5 archs
4	Pi 4 latency reproduced by the posted script	**TBD-on-Pi** — `-demo` mode measures on-board; Pi not yet provisioned
5	Held-out test set, unseen speakers	**done** — 115 test + 5 holdout speakers, disjoint from 307 train
6	Baseline of comparable size compared	**done** — CNN (204k) & MLP (40k) vs Conformer-S (246k), §1

- **GitHub repository**: *to be published (public, MIT)*
- **Dataset location**: Hugging Face, DOI 10.57967/hf/10723 (open access)
- **Compute cluster**: local Apple-Silicon MPS (no A100 available); 1 node, 84 epochs, seed 0
- **Model weights**: `pi_bundle/models/` (ONNX float + int8) · MIT